Tayte Choudhury

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EDUCATION

Texas A&M University — **College Station, Texas**
Master of Science in Finance — *May 2028*
Mays Business School — *Cumulative GPA: In Progress*

Texas A&M University — **College Station, Texas**
Bachelor of Science in Computer Science, Minor in Math — *May 2027*
College of Engineering — *Cumulative GPA: 3.60/4.00*

PROFESSIONAL EXPERIENCE

Ryan LLC — **Houston, Texas**
Severance Tax and Software Engineering Intern — *May 2026 – August 2026*

- Developed **11 production automation tools** using PowerShell, JavaScript, REST APIs, and SSIS to streamline severance tax research, regulatory document retrieval, and automated reporting workflows, saving **60–72 hours of manual work annually**
- Built batch processing tools to retrieve **100s of regulatory filings** and search **100K+ PDF pages** that generate an automated Excel report spanning **6 categories** with data validation, caching, and allowing resumable processing for the entire firm
- Automated a **7-step RRC pipeline** for **52 annual runs**, enabling **1-click** data migration into Ryan's SQL Server databases

Switchless — **College Station, Texas**
Hardware and Software Engineer — *January 2025 – May 2025*

- Contributed to hardware and firmware development, presenting to angel investors and securing **$300K in seed funding** stage
- Designed **4 EasyEDA schematics** for smart switches in new Houston homes, enabling AI automations for **1,000+ users**
- Built embedded-C modules for switch control, signal processing, and optimizing responsiveness across all deployed switches

TanChes — **Houston, Texas**
Front Desk Level One Helpdesk — *May 2023 – August 2023*

- Delivered Level 1 tech support to **100+ users**, resolving hardware, software, and network issues; handled **20–30 tickets** daily
- Deployed, cabled, and configured **14 servers/network devices** for data center operations, securing system reliability/uptime
- Authored technical documentation, including troubleshooting guides and setup procedures to enhance workflow efficiency

PROJECTS

Fused Softmax CUDA Kernel | *CUDA, C++, PyTorch, Nsight Compute* — August 2026

- Implemented single-pass online **softmax** on an RTX 3070 (sm_86), fusing max reduction, exponentiation, and normalization
- Achieved **381 GB/s** effective bandwidth (**85% of 448 GB/s** peak) across **5** tensor shapes spanning **128–65,536** columns
- Optimized SGEMM through **3** stages (naive, shared-memory tiled, register-blocked) to achieve **7,840 GFLOP/s**, **39%** of the **20.3 TFLOP/s** FP32 peak on **2048^3** matrices, a measured 31x gain over the naive kernel and **58%** of cuBLAS throughput

Arbitrage Free Volatility Surface | *Python, SciPy, NumPy* — June 2026 – July 2026

- Built implied volatility surfaces from **4,180 SPY option** quotes spanning 18 expiries and 240 strikes, filtering **31%** of raw quotes through 5 documented liquidity screens and extracting forward prices via put-call parity regression across **18 slices**
- Inverted **2,880 Black-76** prices with Brent's method to 1e-6 convergence tolerance, discarding **62 non-invertible quotes**
- Calibrated 5-parameter SVI slices under 4 no-arbitrage constraints to **0.38 vol-point RMSE**, verified Durrleman butterfly conditions on a **2,000-point moneyness grid** and calendar monotonicity across **17 expiry pairs**, correcting 3 violations

Order Flow Imbalance in Limit Order Books | *Python, pandas, NumPy, statsmodels* — April 2026

- Reconstructed top-of-book state from **347,000** NASDAQ ITCH message events across a full trading session, aligning 6-column message and 40-column orderbook snapshot files and filtering out **12,400** auction, cross, and trading halt records
- Measured bid-ask size imbalance against forward mid-price changes at **10/50/100** event horizons across **10** imbalance deciles
- Found a **0.8 bp** edge in the top decile against a **1.2 cent** median half-spread with Newey-West cutting the naive t-statistic **4.2x**

Aggie Lemons Racing Gear Shifter | *Embedded C, Arduino, EasyEDA* — May 2024 – August 2025

- Built a low-power gear shift indicator using transistors and LEDs, improving driver lap time and cutting power use by **~60%**
- Integrated system into the race car's harness and tested in real time with drivers improving lap times by up to **0.3 seconds**
- Developed custom Arduino software with driver-specific RPM thresholds for LED ignitions, improving usability across users

SKILLS, ACTIVITIES & INTERESTS

Skills: C/C++, CUDA, Python (NumPy, pandas, SciPy, statsmodels, PyTorch), Java, SQL, R, PowerShell, JavaScript, Fusion360
Activities: Aggie Investment Club, Aggie Lemons, Society of Asian Scientists and Engineers, Intramural Sports, The Big Event
Interests: Momentum Analysis, Soccer, Baduk, Sudoku, Game Theory, Formula 1 Load Transfer Analysis, Čzerný's Etudes
Languages: Korean (Fluent), Spanish (Elementary)